



THE CYBERSECURITY ROADMAP

Phase 1: Building the Foundations

01

Networking

- ▶ OSI & TCP/IP Models
- ▶ IP Addressing & Subnetting
- ▶ DNS, DHCP & NAT
- ▶ Firewalls, VPNs & Proxies
- ▶ Packet Analysis (Wireshark)

02

OS & Linux

- ▶ Core CLI Commands
- ▶ File Systems & Permissions
- ▶ User & Process Management
- ▶ System Logs & Monitoring
- ▶ SSH & Remote Access

03

Programming

- ▶ Python for Automation
- ▶ Shell & Bash Scripting
- ▶ Interacting with APIs
- ▶ Data Handling & Parsing
- ▶ Regular Expressions (Regex)

"You cannot secure a system if you do not understand how it operates."



CORE SECURITY SKILLS

Phase 2: Understanding Vulnerabilities

04

Web Security

- ▶ OWASP Top 10 Standards
- ▶ SQL Injection (SQLi)
- ▶ XSS & CSRF Attacks
- ▶ Broken Access Control
- ▶ Security Headers & Configs

05

Cryptography

- ▶ Symmetric vs. Asymmetric
- ▶ Hashing (SHA-256, bcrypt)
- ▶ Public Key Infrastructure (PKI)
- ▶ SSL/TLS Protocols
- ▶ Digital Signatures

06

Vulnerability Mgmt

- ▶ Network Reconnaissance
- ▶ Scanning Tools (Nmap, Nessus)
- ▶ Risk & Threat Modeling
- ▶ CVE Databases Navigation
- ▶ Patch Management Basics

"Think like an attacker. Build like a defender."



CHOOSE YOUR PATH

Phase 3: Industry Specializations

Offensive

Red Team

- ▶ Ethical Hacking & Pen Testing
- ▶ Exploit Development
- ▶ Privilege Escalation
- ▶ Social Engineering
- ▶ Bug Bounty Hunting

Defensive

Blue Team

- ▶ SOC & Incident Response
- ▶ SIEM & Log Analysis (Splunk)
- ▶ Digital Forensics
- ▶ Malware Analysis
- ▶ Threat Intelligence

Architecture

Cloud & AppSec

- ▶ AWS / Azure / GCP Security
- ▶ Identity & Access Mgmt (IAM)
- ▶ DevSecOps & CI/CD Security
- ▶ Kubernetes Security
- ▶ Zero Trust Implementation

"Explore the landscape. Find your niche. Make an impact."



THE FUTURE & MASTERY

Phase 4: Evolving with the Threat Landscape

Next-Gen Trends

- ✓ **AI-Powered Threat Hunting:** Utilizing machine learning to detect anomalies.
- ✓ **Zero Trust Architecture:** "Never trust, always verify" across all perimeters.
- ✓ **Automated Defense:** Self-healing infrastructure and rapid incident response.
- ✓ **Quantum Preparedness:** Preparing for post-quantum cryptographic standards.

Action Plan for Success

- ✓ **Certifications:** Validate skills with CompTIA Security+, CEH, OSCP, or CISSP.
- ✓ **Hands-On Practice:** Engage consistently on platforms like Hack The Box & TryHackMe.
- ✓ **Stay Updated:** Read whitepapers, follow CVE feeds, and join security communities.
- ✓ **Mindset:** Remain curious and ethical.

"Cybersecurity is a journey, not a destination. Keep learning. Keep securing."